1. An integer which is equal to the number of protons in the nucleus of an atom and also equal to the number of electrons in a neutral atom is known as (a) atomic mass (b) neutron number (c) isotopic number (d) atomic number.
2. ....results from sharing one or more electron pairs between two atoms. (a) Covalent bonding (b) ionic bonding (c) hydrogen bonding (d) metallic bonding.
3. All these are characteristics of chemical reaction EXCEPT: (a)evolution of gas (b)change in state (c)formation of precipitate (d) atomic radius.
4. The formation of metallic bond is favoured by the following EXCEPT (a) large atomic radius (b) high ionization energy (c) low ionization energy (d) large number of electrons in the valence shell.
5. Reactions in which two or more substances combine to form a compound are known as.........reactions (a) analysis (b) synthesis (c) combustion (d) replacement.
6. A reaction in which two compounds exchange bonds or ions in order to form different compounds is called (a) double displacement reaction (b) displacement reaction (c) decomposition reaction (d) double decomposition reaction.
7. Our teeth and bones were formed by very slow .... reactions in which mostly calcium phosphate was deposited in the correct geometric arrangements. (a) substitution (b) neutralization (c) gas-formation (d) precipitation.
8. The chemical equation: $Cl_2 + H_2O \rightarrow HCl + HClO$, represents ....reaction. (a) proportionation (b) displacement (c) disproportionation (d) analysis
9. What type of reaction is represented by the following chemical equation? $2NaNO_3(s) \rightarrow 2NaNO_2(s) + O_2(g)$.(a) Double displacement reaction (b) displacement reaction (c) decomposition reaction (d) Double decomposition reaction.
10. One of the following is always a redox reaction. (a) Double displacement reaction (b) displacement reaction (c) decomposition reaction (d) Double decomposition reaction.
11. The removal of ions can occur in three ways in metathesis reactions EXCEPT (a) Formation of predominantly non-ionized molecules (b) Formation of a soluble solid (c) Formation of a precipitate (d) Formation of a gas.
12. One of the following is NOT an attribute of reducing agents (a) reduce other substances (b) contain atoms that are oxidized (c) lose (or appear to lose) electrons (d) contain atoms that are reduced.
13. Reduction in a chemical reaction corresponds to (a) a loss of electrons (b) a gain of electrons (c) addition of oxygen to a compound. (d) increase in oxidation state.
14. .... bonding results from electrostatic interactions among ions, which often results from the net transfer of one or more electrons from one atom or group of atoms to another. (a)Covalent (b) Electrovalent (c) Dative (d) Hydrogen.

50. The following properties distinguishes lithium, Li from other alkali metals EXCEPT (a) Li is soft (b) LiOH is not a strong base (c) LiF is slightly soluble in water (d) Li_2CO_3 evolves carbon dioxide on heating.
51. The group VIIA (group 17) elements are known as the halogens meaning: (a) acid formers (b) salt formers (c) base formers (d) molecule formers.
52. When chlorine and fluorine are combined and heated, they react to form chlorine fluoride, ClF, a compound known as (a) an intrahalogen (b) a mixed halogen (c) an interhalogen (d) disproportionated halogen.
53. On the basis of his mathematical analysis of X-ray data, H. G. J. Moseley concluded that each element differs from the preceding element by having ....more positive charge in its nucleus. (a) one (b) two (c) five (d) three.
54. The predominant isotope of hydrogen does not contain (a) electron (b) proton (c) positron (d) neutron.
55. The number of atoms in each molecule of an element is called (a) atomic chain (b) atomic molecule (c) atomicity (d) molecularity.
56. One of the following pairs are monatomic (a) Helium and Hydrogen (b)Argon and Krypton (c) Arsenic and Xenon (d) Neon and Carbon.
57. The smallest particle of an element or compound that can have a stable independent existence is called (a) an atom (b) a molecule (c) a substance (d) a mixture.
58. An element X has isotopic masses of 6 and 7. If the relative abundance is 1 and 12.5 respectively, what is the relative atomic mass of X? (a) 6.39 (b) 6.93 (c) 6.40 (d) 9.36.
59. The atomic weight of gallium is 69.72 amu. The masses of the naturally occurring isotopes are 68.9257 amu for ${}^{69}_{31}$Ga and 70.9249 amu for ${}^{71}_{31}$Ga respectively. Calculate the percent abundance of ${}^{69}_{31}$Ga. (a) 40 % (b) 30 % (c) 60 % (d) 70 %.
60. Determine the number electrons and protons in: ${}^{37}_{17}Cl^-$ (a) 18 electrons and 17 protons. (b) 17 electrons and 18 protons. (c) 18 electrons and 20 protons. (d) 20 electrons and 18 protons.
61. Determine the number of protons and neutrons in: ${}^{63}_{29}Cu$ (a) 29 protons and 34 neutrons (b) 34 protons and 29 neutrons (c) 27 protons and 63 neutrons (d) 29 protons and 63 neutrons.
62. ....discovered the nucleus of an atom. (a)J.J Thomson (b) Ernest Rutherford (c) Ernest Dalton (d) Humphrey Davy.
63. Different samples of the same compound always contain its constituent elements in the same proportion by mass is the law of......(a) multiple proportions (b) conservation of matter (c) definite proportions (d) Avogadro.
64. ....determined the charge of the electron by 'oil-drop experiment'. (a)Robert Millikan (b) Ernest Millikan (c) Robert Dalton (d) Humphrey Davy.
65. Aurum and Wolfram are the Latin names of .... and ....respectively. (a) gold and tungsten (b) aluminium and tin (c) Argon and tin (d) Arsenic and gold.
66. The Latin names of potassium and mercury are known as........ and .... .respectively. (a) kalium and hydrargyrum (b) natrium and argentum (c) argentum and kalium (d) natrium and hydrargyrum
67. Who discovered neutron? (a) Ernest Chadwick (b) James Chadwick (c) James Rutherford (d) J.J Thomson.
68. The basis of Rutherford's model of the atom is that atoms consist of very small and highly dense positively charged (a) positrons surrounded by clouds of electrons (b) nuclei

15. The bonding between Rb and Cl is primarily (a) coordinate (b) ionic (c) covalent (d) metallic.
16. What type of chemical bonding primarily exists between nitrogen and oxygen? (a) Coordinate (b) ionic (c) covalent (d) metallic.
17. Na^+ and Cl^- are respectively isoelectronic with one of the following pairs: (a) Ni and Ar (b) Ne and He (c) Ar and N (d) Ne and Ar.
18. One of the following is NOT an ionic compound: (a) Li_2O (b) KNO_3 (d) $CaCl_2$ (d) PCl_3
19. When electron pairs are shared unequally in covalent bonds, they are called (a) polar covalent bonds (b) insoluble covalent bond. (c) dative covalent bond (d) non-polar covalent bond.
20. The formula for the ionic compound that forms between Ca and Br is (a)CaBr (b) Ca_2Br (c) $CaBr_2$ (d) Ca_2Br_2.
21. The electron pair is NOT shared equally between atoms of one of the following molecules. (a) H_2 (b) O_2 (c) N_2 (d) HF.
22. A special type of covalent bond in which two atoms are bonded in such a way that one member of the pair is supplying both electrons that are shared is called (a) polar covalent bonds (b) insoluble covalent bond. (c) dative covalent bond (d) non-polar covalent bond.
23. Dispersion forces, which are also called London forces, usually increase with molar mass because molecules with larger molar mass tend to have more (a) protons (b) electrons (c) protons and neutrons (d) neutrons.
24. One of the following is NOT a property of hydrogen bond. (a) They are directional in nature (b) They are essentially electrostatic (c) They are weaker than Vander Waals forces (d) They are weaker than the forces involved in ionic or covalent bonds.
25. Hydrogen compounds (NH_3, H_2O, and HF) have the highest boiling point, contrary to our expectations and do not follow the trend based on molar mass of the elements in groups 5A, 6A, and 7A respectively due to presence of (a) ionic bond (b) non polar bond (c) covalent bond (d) hydrogen bond.
26. Metallic bonding differs from covalent bonding in that metal atoms do not form (a) discrete electron cloud (b) separate ions (c) separate molecules (d) discrete atoms.
27. One of the following represents the relationship between the wavelength and the frequency (a) $\lambda c = v$ (b) $\lambda v = c$ (c) $\lambda w = c$ (d) $\lambda c = w$.
28. Calculate the energy, in joules, of an individual photon of the wavelengths of ultraviolet light of frequency $2.73 \times 10^{16}\ s^{-1}$ (Plank's constant = 6.626×10^{-34} Js) (a) 4.12×10^{49}(b) 2.43×10^{-50} (c) 1.81×10^{-19} (d) 1.81×10^{-17}.
29. Calculate the frequency of radiation with a wavelength of 8973 Å. ($c = 3 \times 10^{8}\ ms^{-1}$, 1Å = 10^{-10} m)(a) $3.34 \times 10^{14} s^{-1}$ (b) $2.99 \times 10^{-15} s^{-1}$ (c) $1.99 \times 10^{-25} s^{-1}$ (d) $4.34 \times 10^{14} s^{-1}$.
30. One of the following represents de Broglie's equation: (a) $\lambda = h/_{mv}$ (b) $\lambda v = c$ (c) $\lambda w = c$ (d) $\lambda c = h/_{mv}$
31. Calculate the wavelength in meters of an electron traveling at 1.24×10^{7} m/s. The mass of an electron is 9.11×10^{-28} g.(1kg = 1000 g, h = $6.626 \times 10^{-34}\ Kgm^2/s$)(a) 5.87×10^{-11}m (b) 1.37×10^{15}m(c) 1.99×10^{-25}m(d) 7.48×10^{-54} m.
32. The Heisenberg Uncertainty Principle states that it is impossible to determine accurately both the .... and the position of an electron (or any other very small particle) simultaneously. (a) momentum (b) photon (c) orbital (d) wavelength.
33. A region of space in which the probability of finding an electron is high is called (a) a molecular orbital (b) an atomic space (c) an atomic orbital (d) an atomic quantum.

34. The angular momentum (or azimuthal or subsidiary) quantum number, l, designates the…. (a) shape of the region in space that an atom occupies (b) shape of the region in space that an electron occupies (c) main energy level (d) specific orbital within a subshell.
35. The sub-shells s, p, d, f corresponds to subsidiary quantum numbers: (a) 1, 2, 3 4 (b) 0, 1, 2, 3 (c) 2, 4, 6, 8 (d) 0, 1, + ½ , - ½.
36. The subshells s, p, d and f are spectroscopic terms and they mean: (a) s-orbital, principle, d-shell and f-orbital (b) shape, principle, decay and full (c) sharp, principal, diffuse and fundamental (d) shock, p-shell, d-orbital and f-shell, respectively.
37. Which basic rules for filling of electron into the orbitals of an atom states that only two electrons can occupy the same orbital and the two electrons must have opposite spins (a) Hund's rule (b) Aufbau's principle (c) Pauli Exclusion Principle (d) Octet rule.
38. Which of the electronic configuration rules for filling of electron into the orbitals of an atom was violated on the following, $1s^2\ 2s^2\ 2px^2\ 2py^2\ 2pz$: (a) Hund's rule (b) Aufbau's principle (c) Pauli Exclusion Principle (d) Octet rule?
39. Two electrons can occupy the same region (orbital) because when they have opposite spins, their opposite magnetic fields helps to overcome the (a) repulsion of their like charges (b) attraction of their like charges (c) repulsion of their unlike charges (d) attraction of their unlike charges.
40. What is the symbol of element composed of the following subatomic particles: $11p$, $12n$, $11e$? (a) Mg (b) Na (c) Al (d) Ne.
41. How many unpaired electrons are in atom of Ne? (a) 0 (b) 2 (c)1(d) 5.
42. Which element is represented by this electronic configuration: $1s^2\ 2s^2\ 2p^6\ 3s^2\ 3p^3$? (a) silicon (b) phosphorus (c) sulphur (d) argon.
43. The periodic law states that the properties of the elements are periodic functions of their (a) mass numbers (b) atomic numbers (c) neutron numbers (d) isoelectronic numbers.
44. Atomic radii decrease from left to right within a given period due to (a) increasing effective nuclear charge (b) decreasing effective nuclear charge (c) addition of electrons to shells farther from the nucleus (d) repulsion of electrons from the nucleus.
45. The amount of energy absorbed when an electron is added to an isolated gaseous atom to form an ion with a -1 charge is known as (a) electron affinity (b) ionization energy (c) electronegativity (d) elctropositivity.
46. One of the following is NOT true: (a) Simple positively charged ions (cations) are always smaller than the neutral atoms from which they are formed (b) Simple negatively charged ions (anions) are always larger than the neutral atoms from which they are formed (c) The sizes of cations decrease from left to right across a period (d) Simple positively charged ions (cations) are always larger than the neutral atoms from which they are formed.
47. The basic repeating unit of a crystal lattice is called (a) a unit cell (b) cubic unit (c) crystal unit (d) amorphous unit.
48. If the cubic unit cell consists of 8 component atoms, molecules or ions located at the corners of the cube, it is called (a) simple cubic (b) body-centered cubic (c) face - centered cubic (d) base - centered cubic.
49. Sodium amalgam used as a reducing agent is a liquid only when the percentage of sodium is (a) low (b) moderate (c) high (d) moderately mixed with other alkali metals.

surrounded by clouds of electrons (c) neutrons surrounded by clouds of electrons (d) nuclei surrounded by clouds of protons.

69. The atomic weight of lithium is 6.941 amu. The two naturally occurring isotopes of lithium have the following masses: ^{6}Li, 6.01512 amu; ^{7}Li, 7.01600 amu. Calculate the percentage of ^{6}Li in naturally occurring lithium. (a) 5.7 % (b) 7.8 % (c) 7.5 % (d) 92.5 %.
70. Atoms that have the same atomic number but different neutron numbers are known as… (a) isotopies (b) allotropes (c) isotopes (d) allotropies.